

USB Intrusion Detection System (USB-IDS)



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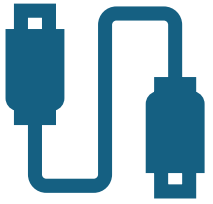
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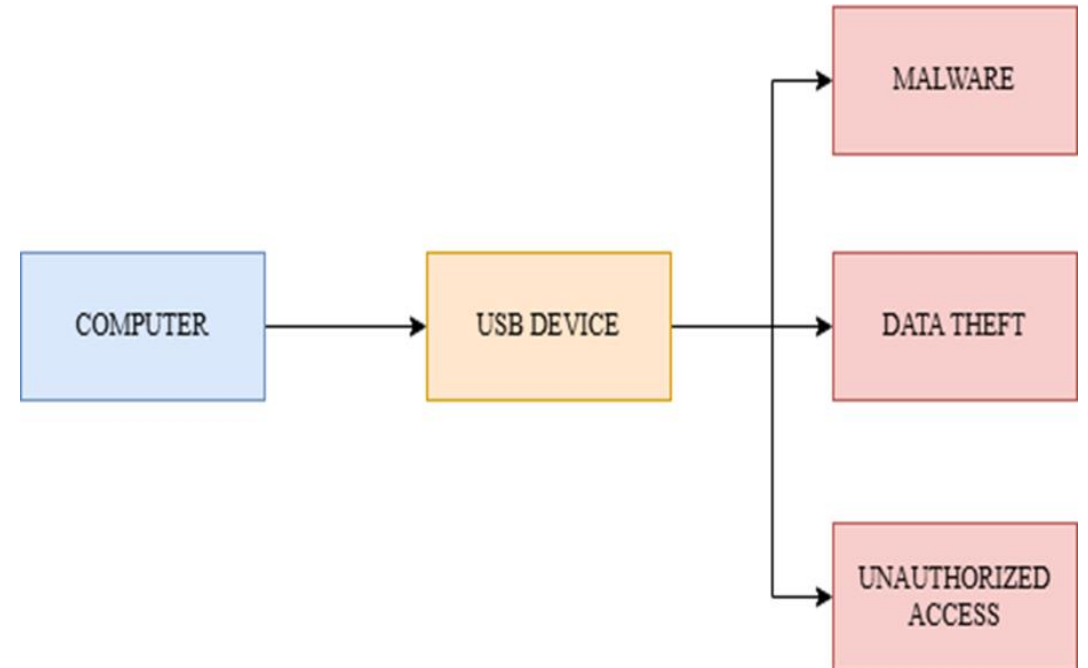
RESULTS & ANALYSIS

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INTRODUCTION

- ✓ Computers are vulnerable to unauthorized USB access
- ✓ USB devices can transfer malware, steal data, or cause system damage
- ✓ To prevent this, USB IDS is developed for:
 - Real-time monitoring of USB events
 - Detecting unauthorized USB devices
 - Alerting users & logging activities
- ✓ Ensures improved system security & access control





Problem Statement

- ✓ Unauthorized USB access can cause data theft & cyber attacks.
- ✓ Need a simple and efficient system to track and control USB usage.



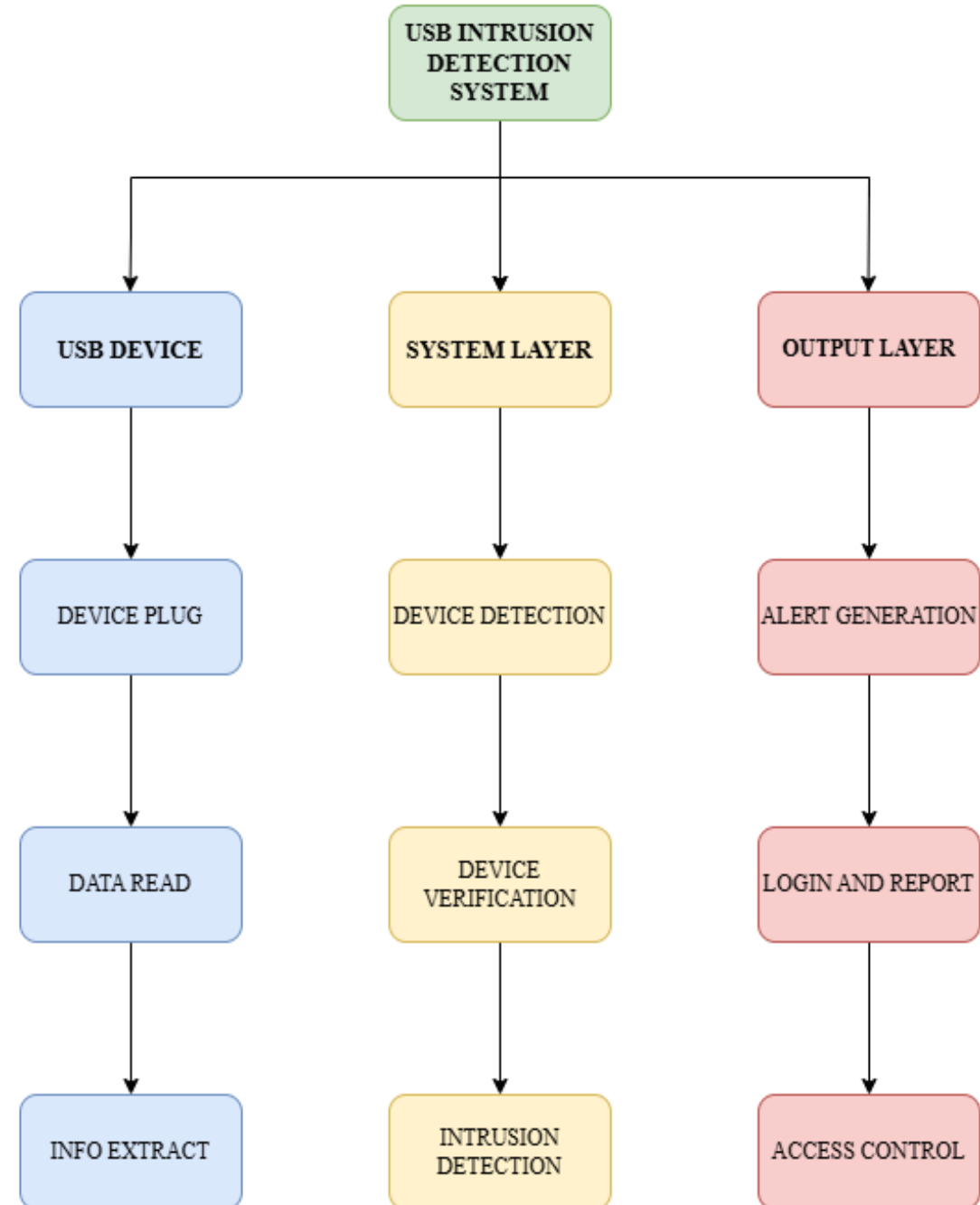
Objectives

- ✓ Monitor USB connections continuously
- ✓ Detect unauthorized USB devices
- ✓ Generate alerts and maintain logs
- ✓ Provide user-friendly interface



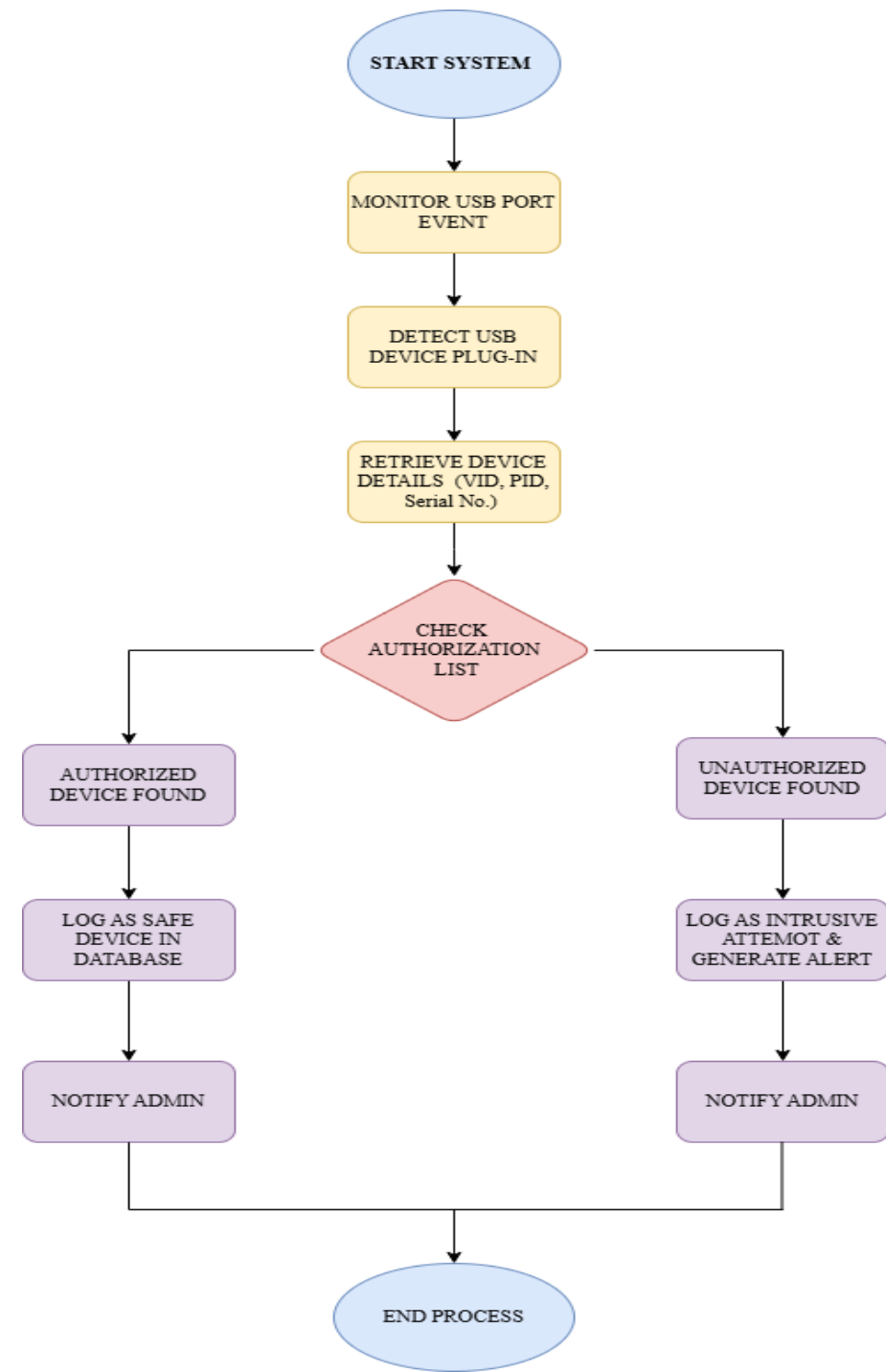
System Analysis & Design

- ✓ Requirement Analysis completed
- ✓ Modules planned for systematic approach:
 - Monitoring Module
 - Detection Module
 - Authorization Module
 - Logging & Alert Module
- ✓ Data Flow Diagram (DFD Level 0 & 1) used to design system structure



Implementation

- ✓ Developed using:
 - Programming: *Python*
 - Database: *SQLite / MySQL*
- ✓ Real-time event tracking
- ✓ Certified device verification using authorized list
- ✓ Alert + Log generation for unauthorized devices



Results & Analysis

- ✓ Dashboard interface to view USB status
- ✓ Pop-up alerts for unauthorized device
- ✓ USB event logs stored in database
- ✓ Improved security compared to normal systems

USB Event Logs

Timestamp	Device Name	Status
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USB IDS Dashboard

No USB Device Connected

Logs Appears Here

Authorized USB Device Connected

Device: SanDisk | Status: Authorized

Event Logged Successfully

Unauthorized USB Device Connected

Device: Unknown | Status: ⚠ ALERT

User Warning: Disconnect Immediately!

CONCLUSION

- ✓ USB IDS successfully detects unauthorized USB devices
- ✓ Helps prevent data theft & improves cyber hygiene
- ✓ Simple, lightweight & suitable for campus & personal systems

FUTURE SCOPE

- ✓ Auto blocking of unauthorized USB
- ✓ Cloud-based centralized monitoring
- ✓ Machine learning-based anomaly detection





THANK
YOU!

